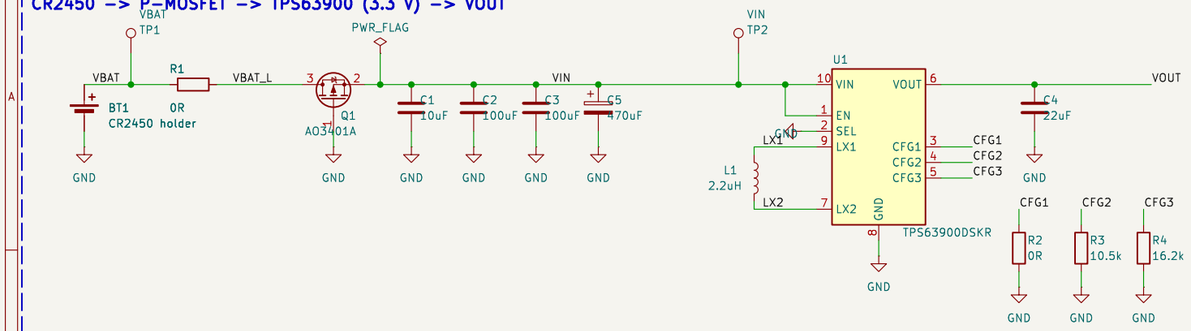
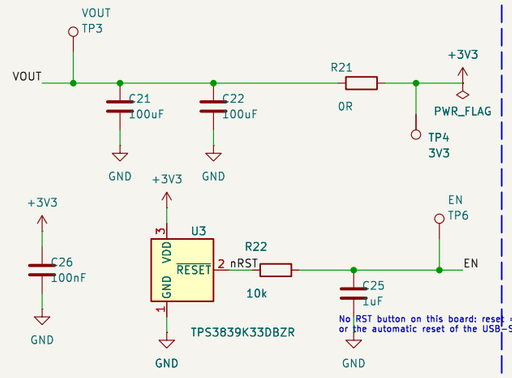
CR2450 -> P-MOSFET -> TPS63900 (3.3 V) -> VOUT



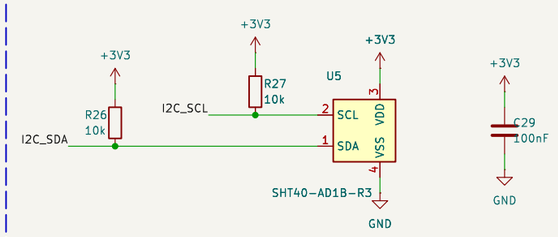
SEL = L: VO(1) = 3.3 V (RCFG3 16.2k, table 7-3). Input current limit 100 mA from RCFG1 0R / RCFG2 10.5k (table 7-2: 13.3k = 50 mA option). TI SLVSET3D, +-1 % resistors. C5: 470 uF electrolytic buffer.

3V3 RAIL, SUPERVISOR, RESET



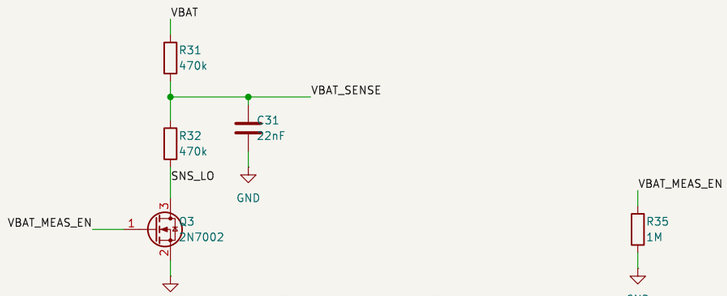
No RST button on this board: reset = short TP6 (EN) to GND, or the automatic reset of the USB-Serial/JTAG port.

SHT40 (on a slotted tongue, away from converters and module)



No copper under the sensor except its pads; do not solder the die pad (SHT4x datasheet).

CELL VOLTAGE SENSE (switched 470k/470k divider, ADC1_CH4 on GPIO5)



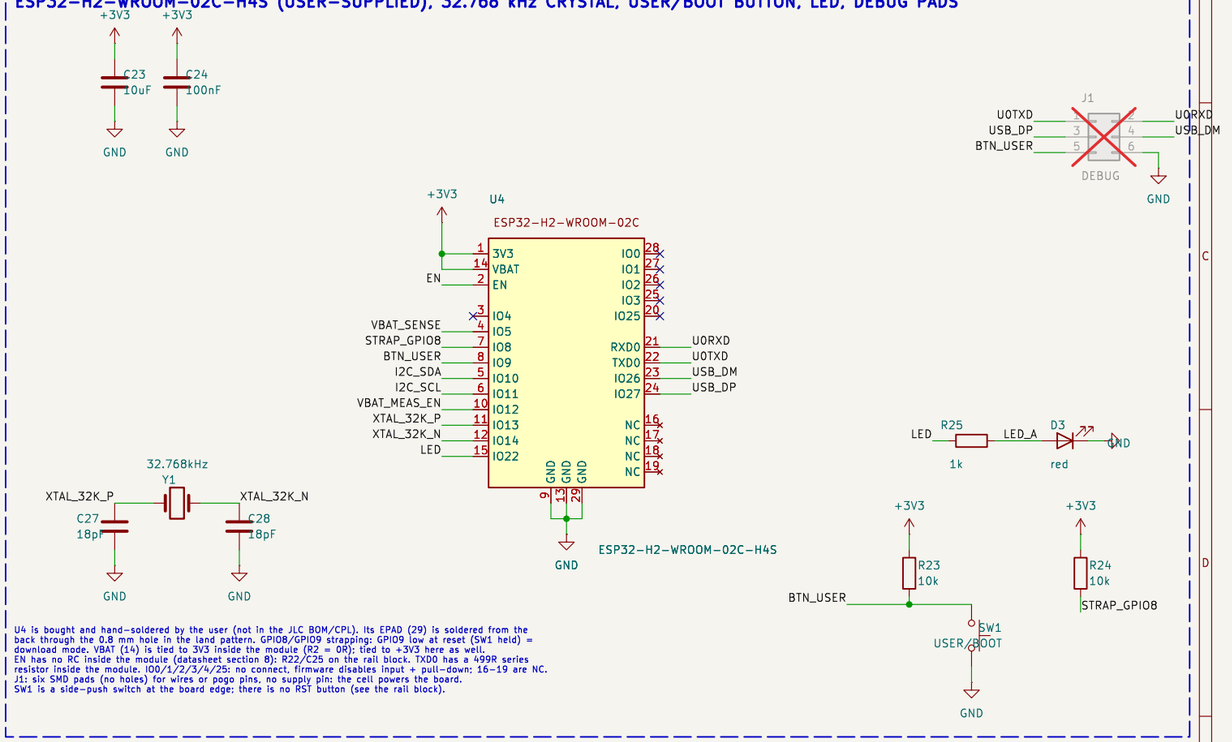
Switch open: no divider current. GPIO12 high for ~30 ms every 6 h: 470k/470k -> ~3.2 uA while on.
The divider top is at the holder terminal (before the 0R link), so a missing cell reads ~0 V.

TEST PADS, MECHANICAL, NOTES



CR2450 ONLY — NOT RECHARGEABLE. No other supply can reach the rail: the cell is the only energy source.
Power-path decision and its limits: docs/decisions.md, docs/power_budget.md (ESTIMATE until measured).
Current measurement: remove R1 (cell) or R21 (load) and insert the ammeter.
U3 TPS3839K33: 2.93 V typ (2.857-2.974 V), push-pull ~RESET, 200 ms delay; R22 keeps it from fighting SW2 / TP6 (EN).

ESP32-H2-WROOM-02C-H4S (USER-SUPPLIED), 32.768 kHz CRYSTAL, USER/BOOT BUTTON, LED, DEBUG PADS



U4 is bought and hand-soldered by the user (not in the JLC BOM/CPL). Its EPAD (29) is soldered from the back through the 0.8 mm hole in the land pattern. GPIO8/GPIO9 strapping: GPIO9 low at reset (SW1 held) = download mode. VBAT (14) is tied to 3V3 inside the module (R2 = 0R); tied to +3V3 here as well.
EN has no RC inside the module (datasheet section 8): R22/C25 on the rail block. TXD0 has a 499R series resistor inside the module. IO0/1/2/3/4/25: no connect, firmware disables input + pull-down; 16-19 are NC.
J1: six SMD pads (no holes) for wires or pogo pins, no supply pin: the cell powers the board.
SW1 is a side-push switch at the board edge; there is no RST button (see the rail block).

Generated by hardware/scripts/gen_schematic.py from design.py — edit there
ERC/DRC reports: hardware/wroom/reports
Power path and battery life figures are ESTIMATES (docs/power_budget.md)
ESP32-H2-WROOM-02C (user-supplied), Matter over Thread, SHT40, 1 x CR2450, TPS63900 3.3 V

TH-SENSOR-H2W

Sheet: /
File: th_sensor_h2w.kicad_sch

Title: TH-SENSOR-H2W thermo-hygrometer (ESP32-H2-WROOM-02C module

Size: A3	Date: 2026-11-08	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 1/1